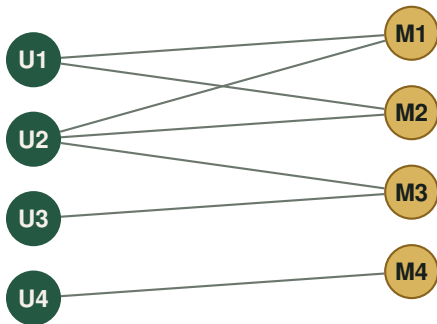
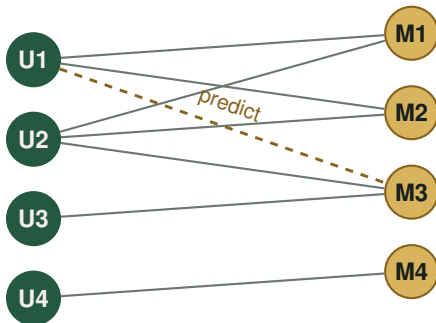


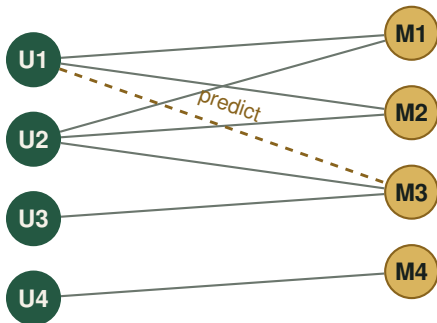
Collaborative filtering



Collaborative filtering



Collaborative filtering



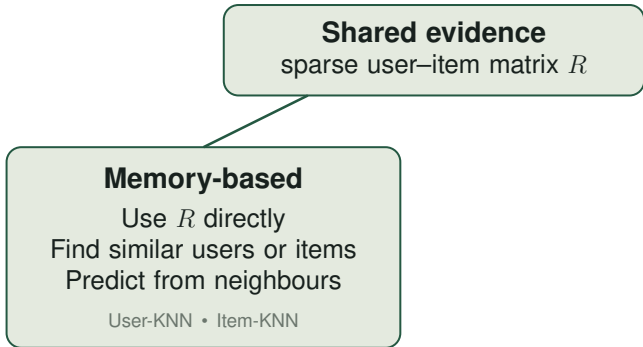
Collaborative signal

If users with similar histories liked M_3 , estimate that U_1 may also like M_3 .

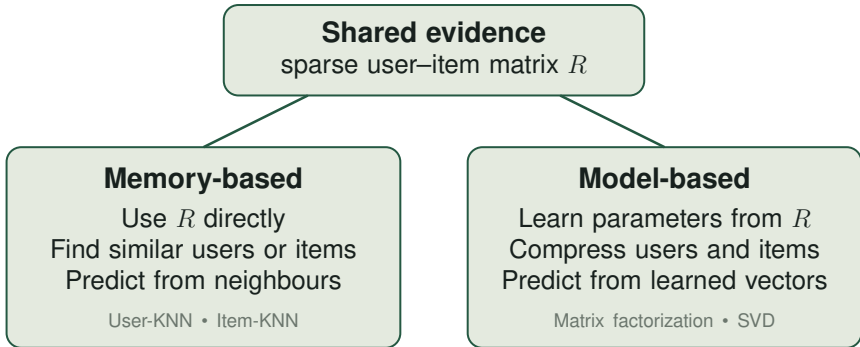
Collaborative filtering has two main families

Shared evidence
sparse user–item matrix R

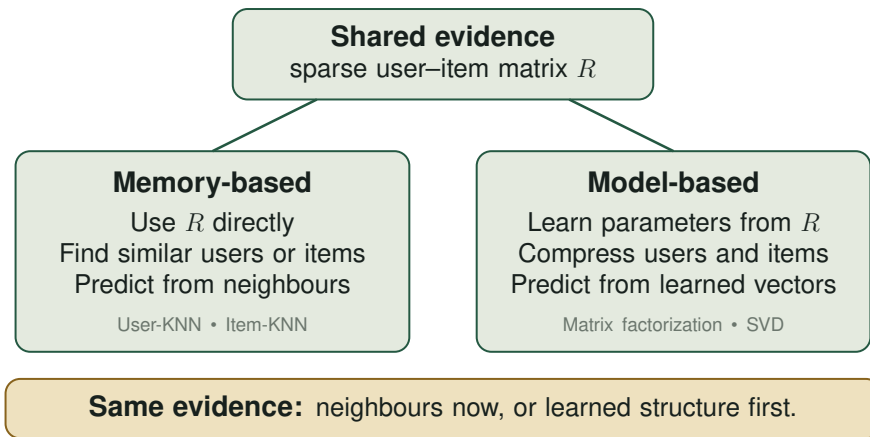
Collaborative filtering has two main families



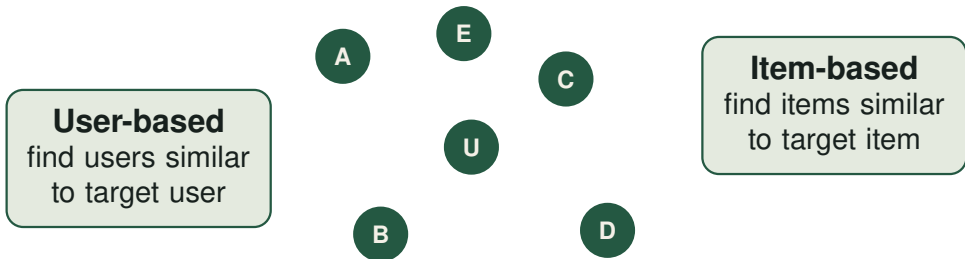
Collaborative filtering has two main families



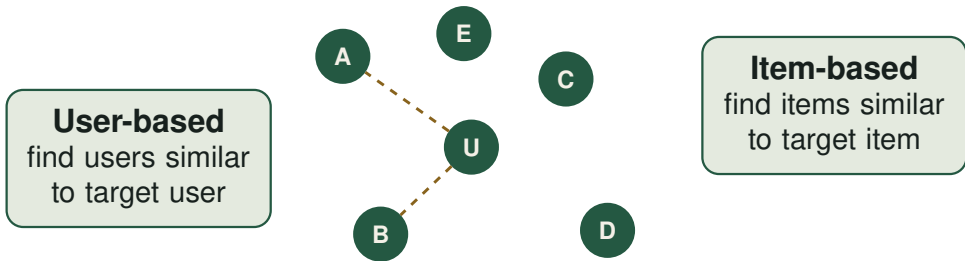
Collaborative filtering has two main families



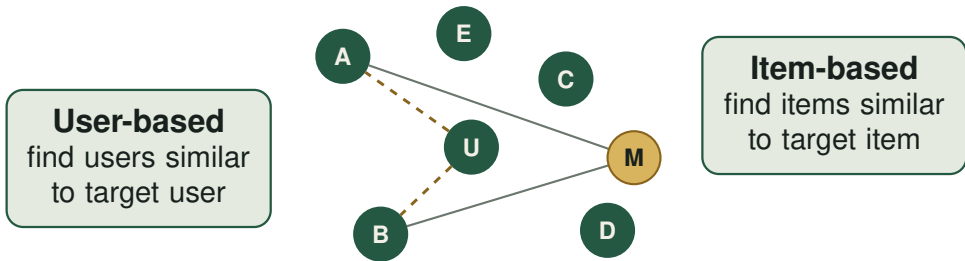
Memory-based methods



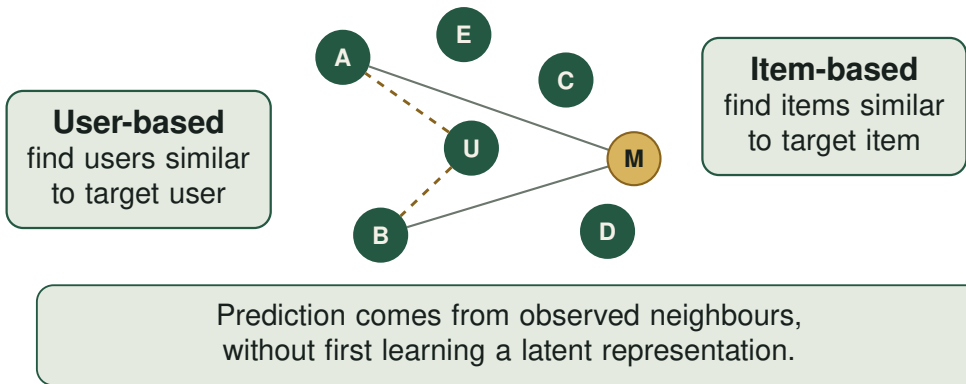
Memory-based methods



Memory-based methods



Memory-based methods



Why memory-based methods struggle at scale

Sparsity

two users may
overlap on
very few items

Why memory-based methods struggle at scale

Sparsity

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Scalability

similarity over
millions of
users/items
is expensive

Why memory-based methods struggle at scale

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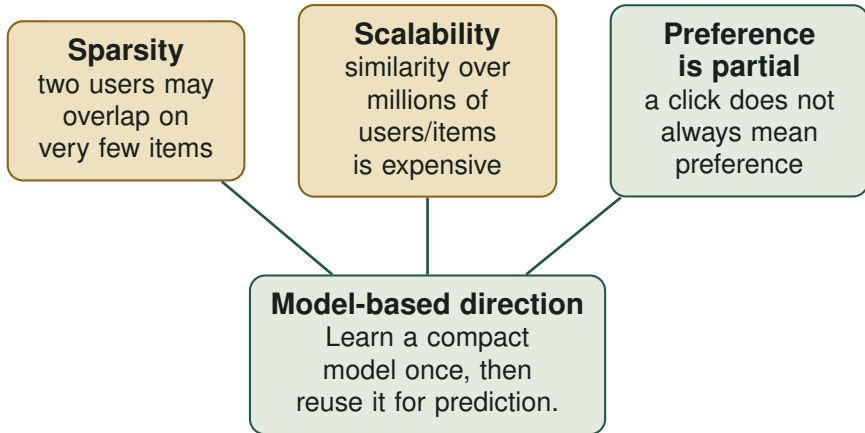
Scalability

similarity over
millions of
users/items
is expensive

Preference is partial

a click does not
always mean
preference

Why memory-based methods struggle at scale



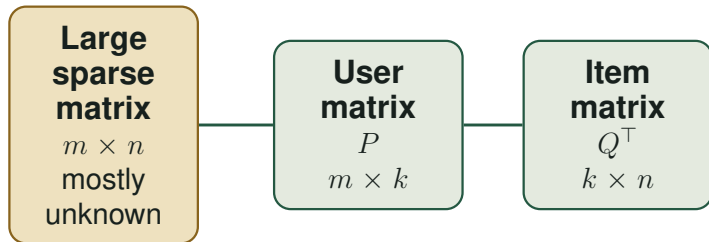
Model-based latent features

**Large
sparse
matrix**

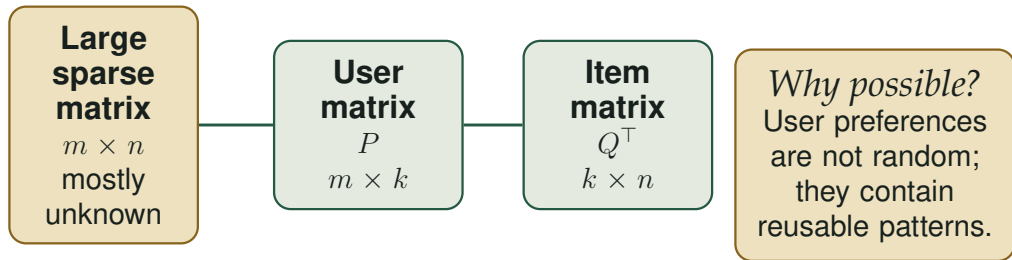
$m \times n$

mostly
unknown

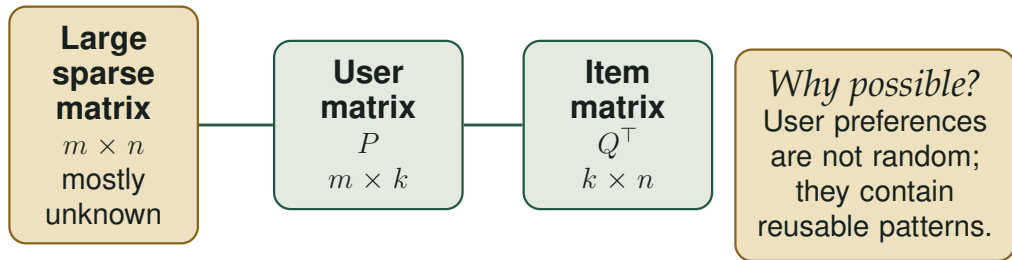
Model-based latent features



Model-based latent features



Model-based latent features



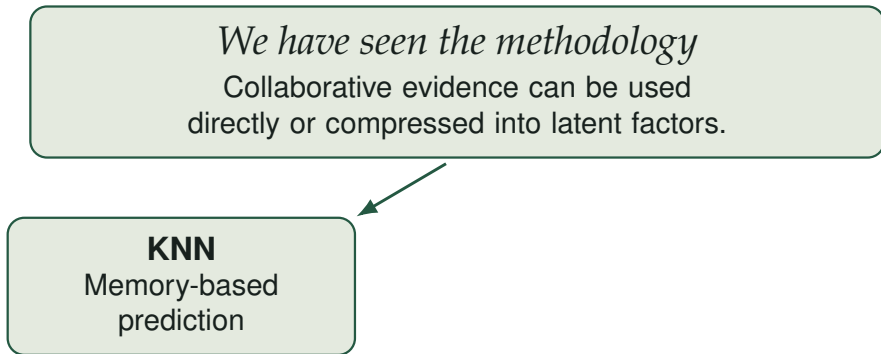
The model may discover axes such as “mainstream vs niche” or “light comedy vs dark drama” without human labels.

How do these methods work in practice?

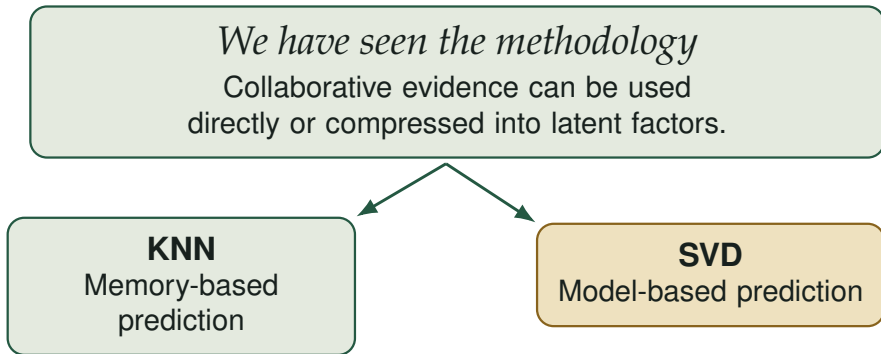
We have seen the methodology

Collaborative evidence can be used
directly or compressed into latent factors.

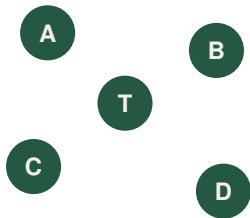
How do these methods work in practice?



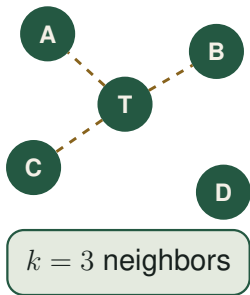
How do these methods work in practice?



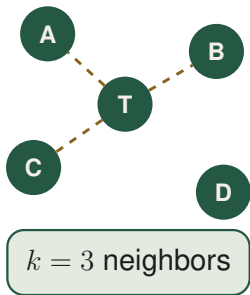
KNN predicts from nearest neighbours



KNN predicts from nearest neighbours



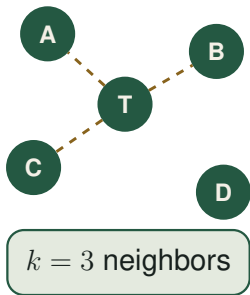
KNN predicts from nearest neighbours



Weighted prediction

$$\hat{r}_{ui} = \bar{r}_u + \frac{\sum_{v \in N(u)} s(u, v)(r_{vi} - \bar{r}_v)}{\sum_{v \in N(u)} |s(u, v)|}$$

KNN predicts from nearest neighbours



Weighted prediction

$$\hat{r}_{ui} = \bar{r}_u + \frac{\sum_{v \in N(u)} s(u, v)(r_{vi} - \bar{r}_v)}{\sum_{v \in N(u)} |s(u, v)|}$$

The neighborhood provides local evidence, usually measured by cosine or Pearson similarity.

SVD learns latent factors

Observed

R

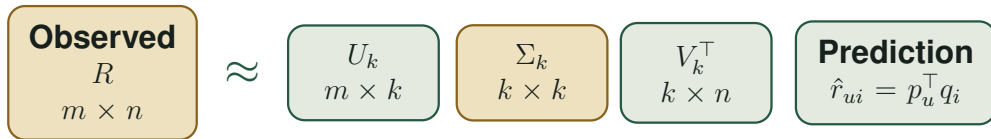
$m \times n$

SVD learns latent factors

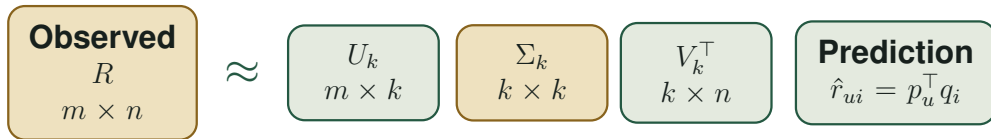
The diagram illustrates the SVD decomposition of an observed matrix R into three latent factor matrices. On the left, a yellow box labeled "Observed" contains the matrix R with dimensions $m \times n$. This is followed by an approximation symbol $\approx$. To the right of the symbol are three boxes: a light green box for U_k with dimensions $m \times k$, a yellow box for Σ_k with dimensions $k \times k$, and a light green box for $V_k^\top$ with dimensions $k \times n$.

$$\begin{array}{c} \text{Observed} \\ R \\ m \times n \end{array} \approx \begin{array}{c} U_k \\ m \times k \end{array} \begin{array}{c} \Sigma_k \\ k \times k \end{array} \begin{array}{c} V_k^\top \\ k \times n \end{array}$$

SVD learns latent factors



SVD learns latent factors



$$\min_{P,Q} \sum_{(u,i) \in \Omega} (r_{ui} - p_u^\top q_i)^2 + \lambda(\|p_u\|^2 + \|q_i\|^2)$$

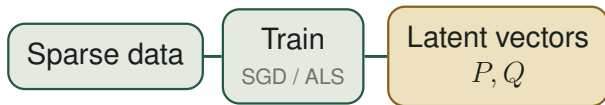
End-to-end recommender workflow

Sparse data

End-to-end recommender workflow



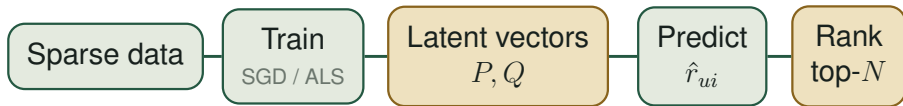
End-to-end recommender workflow



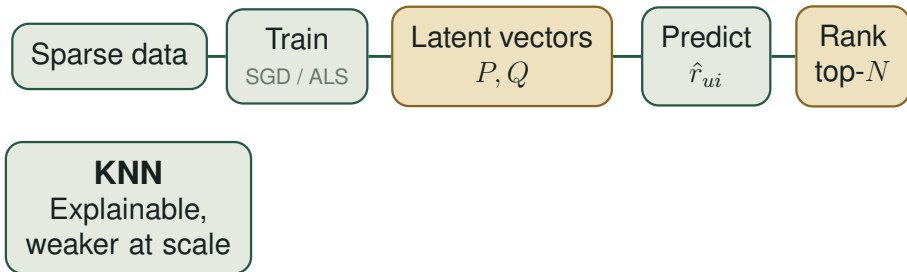
End-to-end recommender workflow



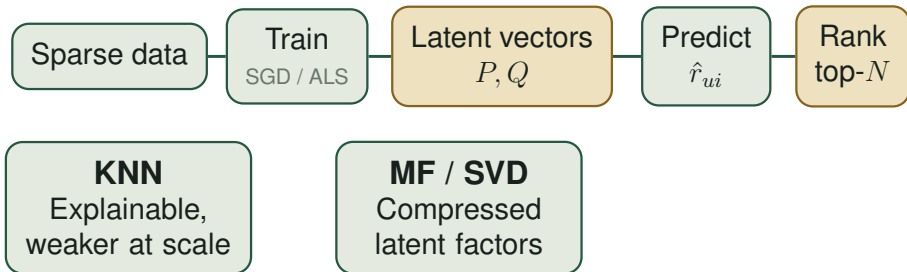
End-to-end recommender workflow



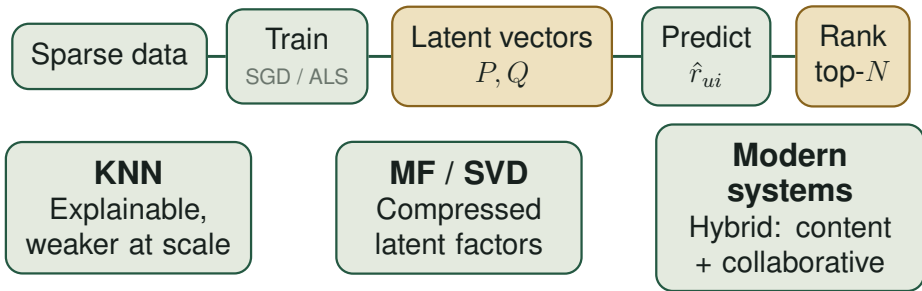
End-to-end recommender workflow



End-to-end recommender workflow



End-to-end recommender workflow



Takeaways

**Sparse
matrix**
Observed
only partially.

Takeaways

Sparse matrix

Observed
only partially.

Two families

Content +
collaborative.

Takeaways

Sparse matrix

Observed only partially.

Two families

Content + collaborative.

Scaling idea

Compress into latent factors.

Takeaways

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Final output

Ranked personalized choices.

Takeaways

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Observed only partially.

Two families

Content + collaborative.

Scaling idea

Compress into latent factors.

Final output

Ranked personalized choices.

**The system does not know exactly what we will like
it estimates from patterns shared across users, items
and context.**